EXPERIMENT 5

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Data Visualization for AI Models Using T5 Translation

AIM:

To implement a Generative AI-based English-to-French text translation system using the T5 Transformer model, evaluate the generated translations using the BLEU score, and visualize the translation performance using different types of plots.

THEORY:

Generative Artificial Intelligence refers to systems capable of generating new content such as text, images, audio, video, and code based on patterns learned from training data. In Natural Language Processing, generative models are used for tasks such as machine translation, text summarization, question answering, and text generation.

In this experiment, the T5-small model is used to perform English-to-French translation. T5 stands for Text-to-Text Transfer Transformer. It is a sequence-to-sequence Transformer model that converts different NLP tasks into a text-to-text format. This means that both the input and the output of the model are represented as text.

1. T5 Transformer Model:

T5 follows an encoder-decoder architecture. The encoder processes the input text and creates a contextual representation. The decoder uses this representation to generate the output sequence one token at a time.

For translation, the input is provided with a task-specific instruction, such as:

```
translate English to French: Hello, how are you?
```

The tokenizer converts the input sentence into numerical token IDs that can be processed by the model. The T5 model then generates the corresponding French translation. The generated token sequence is decoded back into readable text.

2. Tokenization:

Tokenization is the process of converting text into smaller units called tokens. These tokens may represent words, subwords, or punctuation marks. The tokenizer converts the

input sentence into numerical representations that the Transformer model can understand. The generated token IDs are converted back into text after inference.

3. BLEU Score:

BLEU stands for Bilingual Evaluation Understudy. It is a commonly used evaluation metric for machine translation. It compares the machine-generated translation with one or more reference translations created by humans.

BLEU primarily measures the overlap of n-grams between the generated translation and the reference translation. An n-gram is a sequence of n consecutive words. For example, a unigram represents one word, while a bigram represents two consecutive words.

A higher BLEU score generally indicates greater similarity between the generated translation and the reference translation. However, BLEU is based on textual overlap and may not fully capture semantic correctness. A translation can communicate the correct meaning using different words and still receive a lower BLEU score.

4. Data Visualization:

Data visualization is the graphical representation of data that helps identify patterns, distributions, and variations. In this experiment, the calculated BLEU scores are represented using six types of plots:

- Bar Plot: Compares the BLEU scores obtained for different input sentences.
- Line Plot: Displays changes in BLEU scores across sentences.
- Scatter Plot: Shows individual BLEU score values and their distribution.
- Histogram: Represents the frequency distribution of BLEU scores.
- Box Plot: Displays the spread, median, and variation of the scores.
- Area Plot: Shows the BLEU score trend using a filled area.

These visualizations help understand the translation performance of the T5 model across different sentences.

The overall workflow involves loading the pretrained T5 model, tokenizing English input sentences, generating French translations, comparing them with reference translations, calculating BLEU scores, and visualizing the results using Matplotlib.

CODE AND OUTPUT:

Experiment 5 - Data Visualization for AI Models: T5 Translation and BLEU

```
!pip -q install transformers sentencepiece sacrebleu torch matplotlib pandas
```

```
import torch, pandas as pd, matplotlib.pyplot as plt
import sacrebleu
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

MODEL_NAME = "google-t5/t5-small"
device = "cuda" if torch.cuda.is_available() else "cpu"
tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)
model = AutoModelForSeq2SeqLM.from_pretrained(MODEL_NAME).to(device)
model.eval()
print("Model:", MODEL_NAME, "| device:", device)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config.json: 100% 1.21k/1.21k [00:00<00:00, 25.3kB/s]
tokenizer_config.json: 100% 2.32k/2.32k [00:00<00:00, 49.7kB/s]
spiece.model: 100% 792k/792k [00:00<00:00, 9.47MB/s]
tokenizer.json: 100% 1.39M/1.39M [00:00<00:00, 12.2MB/s]
model.safetensors: reconstructing file: 100% 242MB / 242MB, 19.6MB/s
model.safetensors: downloading bytes: 108MB, 8.65MB/s
Loading weights: 100% 131/131 [00:00<00:00, 746.17it/s]
generation_config.json: 100% 147/147 [00:00<00:00, 4.36kB/s]
Model: google-t5/t5-small | device: cpu
```

```
english = [
    "Hello, how are you?",
    "I love artificial intelligence.",
    "The weather is very good today.",
    "I am a computer science student.",
    "Cricket is a popular sport.",
    "The coach gives useful advice."
]
reference = [
    "Bonjour, comment allez-vous ?",
    "J'aime l'intelligence artificielle.",
    "Le temps est très beau aujourd'hui.",
    "Je suis étudiant en informatique.",
    "Le cricket est un sport populaire.",
    "L'entraîneur donne des conseils utiles."
]

translations, scores = [], []
for src, ref in zip(english, reference):
    prompt = "translate English to French: " + src
    inputs = tokenizer(prompt, return_tensors="pt", truncation=True).to(device)
    with torch.no_grad():
        output = model.generate(**inputs, max_new_tokens=48, num_beams=4)
    translated = tokenizer.decode(output[0], skip_special_tokens=True)
    bleu = sacrebleu.sentence_bleu(translated, [ref]).score
    translations.append(translated); scores.append(bleu)
    print("English:", src, "\nReference:", ref, "\nGenerated:", translated, "\nSentence BLEU:", round(bleu,2), "\n")

results = pd.DataFrame({"English":english, "Reference French":reference,
                        "Generated French":translations, "Sentence BLEU":scores})
display(results)
print("Mean sentence BLEU:", round(float(results["Sentence BLEU"].mean()),2))
```

```
English: Hello, how are you?
Reference: Bonjour, comment allez-vous ?
Generated: Bonjour, comment êtes-vous?
Sentence BLEU: 42.73

English: I love artificial intelligence.
Reference: J'aime l'intelligence artificielle.
Generated: J'aime l'intelligence artificielle.
Sentence BLEU: 100.0

English: The weather is very good today.
Reference: Le temps est très beau aujourd'hui.
Generated: Les conditions météorologiques sont très bonnes aujourd'hui.
Sentence BLEU: 12.22
```

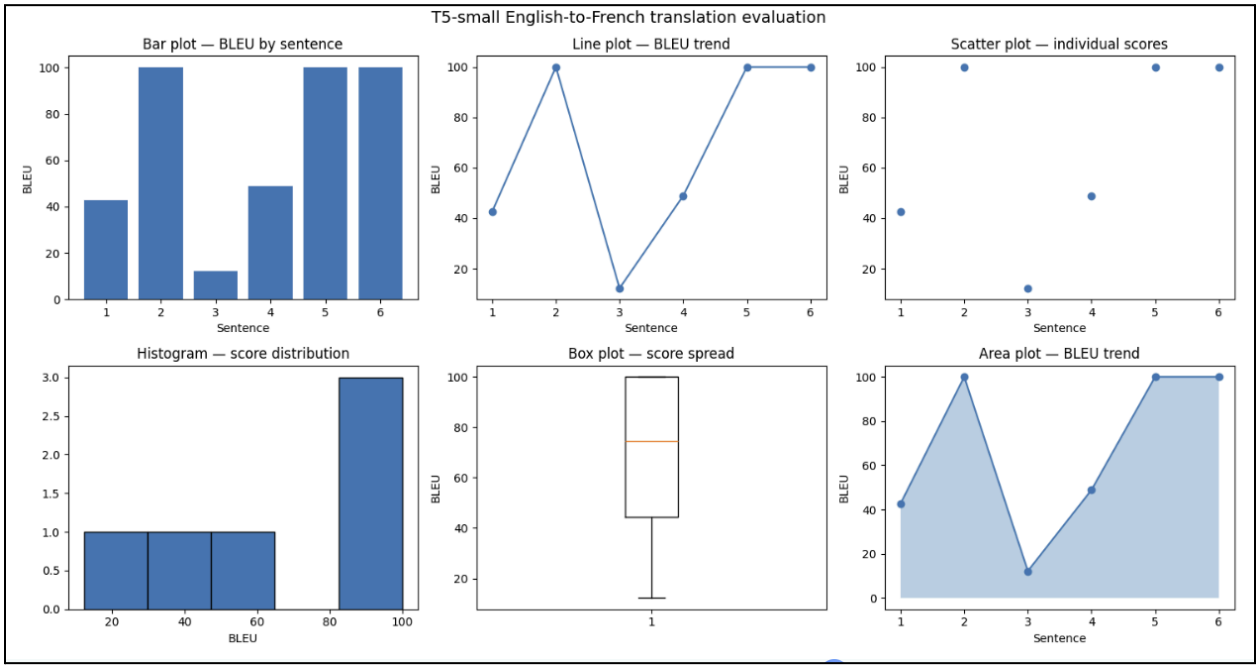
```
English: I am a computer science student.
Reference: Je suis étudiant en informatique.
Generated: Je suis un étudiant en informatique.
Sentence BLEU: 48.89

English: Cricket is a popular sport.
Reference: Le cricket est un sport populaire.
Generated: Le cricket est un sport populaire.
Sentence BLEU: 100.0

English: The coach gives useful advice.
Reference: L'entraîneur donne des conseils utiles.
Generated: L'entraîneur donne des conseils utiles.
Sentence BLEU: 100.0
```

	English	Reference French	Generated French	Sentence BLEU
0	Hello, how are you?	Bonjour, comment allez-vous ?	Bonjour, comment êtes-vous?	42.728701
1	I love artificial intelligence.	J'aime l'intelligence artificielle.	J'aime l'intelligence artificielle.	100.000000
2	The weather is very good today.	Le temps est très beau aujourd'hui.	Les conditions météorologiques sont très bonne...	12.223076
3	I am a computer science student.	Je suis étudiant en informatique.	Je suis un étudiant en informatique.	48.892302
4	Cricket is a popular sport.	Le cricket est un sport populaire.	Le cricket est un sport populaire.	100.000000
5	The coach gives useful advice.	L'entraîneur donne des conseils utiles.	L'entraîneur donne des conseils utiles.	100.000000

```
Mean sentence BLEU: 67.31
```




```
results.to_csv("experiment5_translation_bleu_results.csv", index=False)
print("Saved experiment5_translation_bleu_results.csv")
```

```
Saved experiment5_translation_bleu_results.csv
```

CONCLUSION:

The experiment demonstrated English-to-French translation using the pretrained T5 Transformer model. The generated translations were compared with reference translations using BLEU scores. Six different visualization techniques were used to represent translation performance and score distribution. The experiment helped understand how generative AI models can be evaluated and how visualization makes model performance easier to analyze.